Channel State Information-Free Angle Spoofing for Differential Localization Privacy

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***Abstract*—This paper proposes a channel state information (CSI)-free angle spoofing strategy that remains effective even when the eavesdropper (Eve) uses differential statistics to estimate the location of the transmitter (Alice). Specifically, by designing a precoding matrix, Eve incorrectly estimates the angle differences of departure (ADoDs), thereby spoofing Alice's location. Meanwhile, the trusted user (Bob) can still infer Alice's true location with a modest amount of securely shared side information. Given the channel parameters estimated by Eve from the spoofed channel, the resulting localization bias introduced to the eavesdroppers under the spoofed geometric is analyzed. Eve's bias can be driven to infinity asymptotically in a key parameter of the precoder. Simulation results demonstrate that the proposed method introduces significant localization errors for Eve. When combined with a prior CSI-free spoofing strategy, further degraded performance is achieved by Eve.**

***Index Terms*—Location-privacy, spoofing, precoding, channel state information**

I. Introduction

Due to the large bandwidth and high resolution, millimeter wave (mmWave) signals [1] are widely employed to provide high-accuracy localization. Recent channel estimation algorithms [2], [3], [4] accurately recover multidimensional multipath channel parameters including times of arrivals (ToAs), angles of departure (AoDs) and angles of arrival (AoAs), which can be exploited for localization. While these methods enable accurate positioning, they can also be used by eavesdroppers, allowing unauthorized inference of the transmitter's location.

To combat this issue, physical-layer-based privacy preserving techniques have been investigated. Artificial-noise-based methods enhance physical-layer security by degrading the eavesdropper's channel [5], [6]. Beamforming techniques have also been proposed to hide critical location information from eavesdroppers [7], [8], or limit the eavesdropper's CRB [9]. A power allocation strategy is proposed in [10] to limit position exposure to Eve. However, these methods rely on known CSI.

Location spoofing has emerged as an effective physical-layer strategy for protecting location information. Existing methods, also typically require some CSI knowledge which is generally unavailable since eavesdroppers do not disclose their locations. Angle information of the channel between Eve and Alice is required in [11], [12]. In [13], the direct and reflected channel components are needed to selectively shift the ToA of the direct path. To address this limitation, a CSI-free delay-angle information spoofing (DAIS) method was proposed in [14], [15], [16], where a precoding matrix is designed to shift Eve's ToA and AoD estimates. However, DAIS is less effective in the presence of *differential-signaling-based* localization techniques. Specifically, time differences of arrival (TDoAs) are used in [17], [18], [19], and [20] uses TDoAs, ADoDs and angle differences of arrrival (ADoAs) for distance estimation. As shown herein, DAIS-induced spoofing can fail when using TDoAs and ADoDs.

Therefore, in this paper, we design a precoding-based differential-signaling angle spoofing (DSAS) strategy that can mislead Eve into learning incorrect ADoDs. The main contributions of this paper are as follows:

1) A precoding matrix is designed to spoof Eve's estimates of AoDs as well as the relative AoDs between non-line-of-sight (NLoS) paths and the line-of-sight (LoS) path.
2) The effectiveness of DAIS [14], [15], [16] against differential-signaling localization [20] is analyzed.
3) In the absence of noise, approximate spoofed ADoDs for Eve and Eve's localization error are derived. It is shown that the bias for Eve's estimates can be driven to be large through the careful precoding parameter choice.
4) Simulation results demonstrate the effectiveness of DSAS, and its performance improvement when combined with DAIS under differential localization.

The following notational conventions are adopted: lower case x denotes scalars, a variant of x denotes random scalars, boldface lower case $\boldsymbol{x}$ denotes vectors, a variant of the boldface $\mathbf{x}$ denotes random vectors and boldface upper case $\boldsymbol{X}$ denotes matrices. The (t,k)-th entry of the matrix $\boldsymbol{X}$ is denoted by $X_{t,k}$. We denote $x^\star$ and $\boldsymbol{x}^\star$ as the **ground-truth** values of scalar and vector parameters. The quantities $\hat{x}\left(\cdot\right)$ and $\hat{\boldsymbol{x}}\left(\cdot\right)$ denote estimates of scalar and vector parameters, respectively. The operators $\left(\cdot\right)^\top$, $\overline{\left(\cdot\right)}$, $\left(\cdot\right)^{\mathsf{H}}$, and $\left(\cdot\right)^\dagger$ denote transpose, conjugate, conjugate transpose, and pseudoinverse, respectively. The Hadamard product is given by $\odot$.

II. Preliminaries

A. Wireless Channel Models

We consider a multiple-input multiple-output (MIMO) channel with orthogonal frequency-division multiplexing (OFDM) signaling. The frequency-domain channel over the n-th subcarrier is given by

$$\boldsymbol{H}^{(n)} = \sum_{k=0}^{K} \gamma_k^\star e^{\frac{-j2\pi n \tau_k^\star}{NT_s}} \boldsymbol{A}_{N_R,N_T}\left(\theta_{Rx,k}^\star, \theta_{Tx,k}^\star\right), \quad (1)$$

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for $n \in \{1, \cdots, N\}$, where $K+1$, N_R and N_T represent the numbers of propagation paths, receivers, and transmitters, respectively. The parameters $\gamma_k^\star$, $\tau_k^\star$, $\theta_{Rx,k}^\star$ and $\theta_{Tx,k}^\star$ represent the ground-truth amplitude, ToA, AoA and AoD for the k-th path, where $\tau_0^\star \leq \tau_1^\star \leq \cdots \leq \tau_K^\star$ and the line-of-sight (LoS) path is indexed by 0. For $\vartheta, \varphi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, the steering matrix is defined as

$$\boldsymbol{A}_{N_R,N_T}(\vartheta,\varphi) \doteq \boldsymbol{a}_{N_R}(\vartheta)\, \boldsymbol{a}_{N_T}^{\mathsf{H}}(\varphi) \in \mathbb{C}^{N_R \times N_t}, \quad (2)$$

where the exponential steering vector takes the form

$$\boldsymbol{a}_{N_I}(\vartheta) \doteq \left[1, e^{-j\pi \sin\vartheta}, \cdots, e^{-j\pi(N_0-1)\sin\vartheta}\right]^\top \in \mathbb{C}^{N_I}, \quad (3)$$

for $N_I \in \{N_R, N_T\}$ and $\vartheta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. The distance between antennas is set to half the wavelength.

We assume N independent observation blocks,

$$\mathbf{Y}^{(n)} = \left(\boldsymbol{W}^{\mathsf{H}} \boldsymbol{H}^{(n)} \boldsymbol{F}\right) \odot \boldsymbol{S}^{(n)} + \mathbf{E}^{(n)} \in \mathbb{C}^{R_1 \times R_2}, \quad (4)$$

where the noise $\mathrm{E}_{r_1,r_2}^{(n)} \overset{\text{i.i.d}}{\sim} \mathcal{CN}\left(0, \sigma_e^2\right)$ and $\boldsymbol{S}^{(n)} \in \mathbb{C}^{R_1 \times R_2}$ is the n-th transmitted symbol block. $\boldsymbol{W} \in \mathbb{C}^{N_R \times R_1}$ and $\boldsymbol{F} \in \mathbb{C}^{N_T \times R_2}$ are the combiner and precoder matrices, respectively.

B. Differential-Signaling Localization Method

Suppose an authorized user (Alice) is at the location $\boldsymbol{p}^\star \in \mathbb{R}^2$. A localizer (Bob/Eve) at $\boldsymbol{q}^\star \in \mathbb{R}^2$ is trying to infer the location of Alice, with geometric parameters $\boldsymbol{\tau} = [\tau_0, \cdots, \tau_K]^\top$, $\boldsymbol{\theta}_{Rx} = [\theta_{Rx,0}, \cdots, \theta_{Rx,K}]^\top$ and $\boldsymbol{\theta}_{Tx} = [\theta_{Tx,0}, \cdots, \theta_{Tx,K}]^\top$. We define the corresponding *differential parameters* as $\dot{\boldsymbol{x}} \doteq [|x_1 - x_0|, \cdots, |x_K - x_0|]^\top \in \mathbb{R}^K$ for $\boldsymbol{x} \in \{\boldsymbol{\tau}, \boldsymbol{\theta}_{Rx}, \boldsymbol{\theta}_{Tx}\}$.

According to the differential-signaling localization method [20], the localizer estimates Alice's location as

$$\boldsymbol{p} = \boldsymbol{q}^\star - d\,[\cos\theta_{Rx,0}, \sin\theta_{Rx,0}]^\top \in \mathbb{R}^2, \quad (5)$$

where d is given by,

$$d = \frac{\left\langle c\dot{\boldsymbol{\tau}} \odot \sin\left(\dot{\boldsymbol{\theta}}_{Rx} + \dot{\boldsymbol{\theta}}_{Tx}\right), \boldsymbol{f}\left(\dot{\boldsymbol{\theta}}_{Rx}, \dot{\boldsymbol{\theta}}_{Tx}\right)\right\rangle}{\left\|\boldsymbol{f}\left(\dot{\boldsymbol{\theta}}_{Rx}, \dot{\boldsymbol{\theta}}_{Tx}\right)\right\|_2^2}, \quad (6)$$

and the function $f : \mathbb{R}^l \times \mathbb{R}^l \to \mathbb{R}^l$ is defined as

$$\boldsymbol{f}(\boldsymbol{\vartheta}, \boldsymbol{\varphi}) \doteq \sin\boldsymbol{\vartheta} + \sin\boldsymbol{\varphi} - \sin(\boldsymbol{\vartheta} + \boldsymbol{\varphi}), \quad (7)$$

where $\sin(\cdot)$ is element-wise for $\boldsymbol{\vartheta}, \boldsymbol{\varphi} \in \mathbb{R}^l$.

Notice that Equation (6) can be expressed as the weighted average over the pairwise distance estimates

$$d = \sum_{k=1}^K c_k \frac{c\dot{\tau}_k \sin\left(\dot{\theta}_{Rx,k} + \dot{\theta}_{Tx,k}\right)}{f\left(\dot{\theta}_{Rx,k}, \dot{\theta}_{Tx,k}\right)} \doteq \sum_{k=1}^K c_k d_k, \quad (8)$$

where d_k is the distance estimate with the k-th LoS-NLoS pair and $c_k = \frac{f^2\left(\dot{\theta}_{Rx,k}, \dot{\theta}_{Tx,k}\right)}{\sum_{k'=1}^K f^2\left(\dot{\theta}_{Rx,k'}, \dot{\theta}_{Tx,k'}\right)} \geq 0$ is the weight.

Using trigonometric equalities, we can further simplify the expression for d_k:

$$d_k \overset{(a)}{=} \frac{c\dot{\tau}_k \cdot 2\sin\frac{\dot{\theta}_{Rx,k}+\dot{\theta}_{Tx,k}}{2}\cos\frac{\dot{\theta}_{Rx,k}+\dot{\theta}_{Tx,k}}{2}}{2\sin\frac{\dot{\theta}_{Rx,k}+\dot{\theta}_{Tx,k}}{2}\left(\cos\frac{\dot{\theta}_{Rx,k}-\dot{\theta}_{Tx,k}}{2} - \cos\frac{\dot{\theta}_{Rx,k}+\dot{\theta}_{Tx,k}}{2}\right)}$$

$$\overset{(b)}{=} \frac{c\dot{\tau}_k}{2}\left(\cot\frac{\dot{\theta}_{Rx,k}}{2}\cot\frac{\dot{\theta}_{Tx,k}}{2} - 1\right), \quad (9)$$

where (a) follows by $\sin\vartheta + \sin\varphi = 2\sin\frac{\vartheta+\varphi}{2}\cos\frac{\vartheta-\varphi}{2}$ and $\sin(2\vartheta) = 2\sin\vartheta\cos\vartheta$, for $\vartheta, \varphi \in \mathbb{R}$. (b) follows by $\cos(\vartheta+\varphi) = \cos\vartheta\cos\varphi - \sin\vartheta\sin\varphi$ and $\cos\vartheta - \cos\varphi = 2\sin\frac{\vartheta+\varphi}{2}\sin\frac{\varphi-\vartheta}{2}$, for $\vartheta, \varphi \in \mathbb{R}$.

C. Potential Failure of DAIS for Eve

Recall that DAIS [16] designs the precoding matrix as

$$\boldsymbol{\Phi}^{(n)}(\Delta_\tau, \Delta_\theta) = e^{-\frac{j2\pi n\Delta_\tau}{NT_s}} \mathrm{diag}\left\{a_{N_T}^{\mathsf{H}}(\Delta_\theta)\right\}, \quad (10)$$

for $\Delta_\tau \in (0, NT_s]$ and $\Delta_\theta \in (0, \pi]$.

The spoofing effect will be eliminated in the presence of the above differential signaling, unless **one** of the following conditions are satisfied: (i) $\tau_0 + \Delta_\tau < NT_s$ and $\tau_K + \Delta_\tau > NT_s$; (ii) $\min_{k=0,\cdots,K} \theta_{Tx,k} + \Delta_\theta < \frac{\pi}{2}$ and $\max_{k=0,\cdots,K} \theta_{Tx,k} + \Delta_\theta > \frac{\pi}{2}$.

From our numerical results in the sequel, we will see that there are channel realizations for which DAIS fails under the above differential-signaling localization, whereas DSAS always works. Moreover, the probability that Eve is in such a position increases as the number of paths decreases. However, DAIS can still introduce significant errors into Eve's estimates for certain channel realizations – it does not always fail in differential environments.

III. Differential-Signaling Angle Spoofing

Our aim is to design a precoding strategy inspired by [16] that yields a spoofing of Alice's position that is robust to the use of differential-signaling localization [20]. Furthermore, the spoofing effect should be achieved without Alice possessing any information about the CSI from herself to the localizer.

We observe that $\boldsymbol{A}_{N_R,N_T}(\vartheta_1,\varphi_1)\,\boldsymbol{a}_{N_T}(\vartheta_2)\,\boldsymbol{a}_{N_T}^{\mathsf{H}}(\varphi_2) = \left[\boldsymbol{a}_{N_T}^{\mathsf{H}}(\varphi_1)\,\boldsymbol{a}_{N_T}(\vartheta_2)\right]\boldsymbol{A}_{N_R,N_T}(\vartheta_1,\varphi_2)$, where the parameter φ_1 of the steering matrix is spoofed to an arbitrary angle φ_2. Motivated by this, we design the following precoding matrix to spoof the AoDs of the channel:

$$\boldsymbol{\Psi}\left(\boldsymbol{\phi}_{Tx}^{(n)}\right) = \sum_{m=0}^{M-1} \boldsymbol{B}_{N_T}\left(\sin^{-1}\left(\frac{2m-M}{M}\right), \phi_{Tx,m}^{(n)}\right), \quad (11)$$

where $M \geq N_T$ is a pre-defined parameter and the matrix $\boldsymbol{B}_{N_T}(\vartheta,\varphi)$ is defined as $\boldsymbol{B}_{N_T}(\vartheta,\varphi) \doteq \boldsymbol{a}_{N_T}(\vartheta)\,\boldsymbol{a}_{N_T}(\varphi)^{\mathsf{H}}$, $\forall \vartheta, \varphi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. The side information which must be shared secretly with Bob is $\boldsymbol{\phi}_{Tx}^{(n)}$. We underscore that $\boldsymbol{\phi}_{Tx}^{(n)}$ is **not** a function of either $\boldsymbol{\theta}_{Rx}^\star$ or $\boldsymbol{\theta}_{Tx}^\star$. With precoding, the n-th observation block at Eve is now

$$\widetilde{\mathbf{Y}}^{(n)} = \left(\boldsymbol{W}^{\mathsf{H}} \boldsymbol{H}^{(n)} \boldsymbol{\Psi}\left(\boldsymbol{\phi}_{Tx}^{(n)}\right) \boldsymbol{F}\right) \odot \boldsymbol{S}^{(n)} + \mathbf{E}^{(n)}. \quad (12)$$

We allow the eavesdropper, Eve, to have significant side information, that is: $\boldsymbol{W}$, $\boldsymbol{F}$ and $\boldsymbol{S}^{(n)}$. Then, the channel experienced by Eve is the spoofed channel given by,

$$\widetilde{\boldsymbol{H}}^{(n)} = \boldsymbol{H}^{(n)} \boldsymbol{\Psi}\left(\boldsymbol{\phi}_{Tx}^{(n)}\right)$$

$$= \sum_{m=0}^{M-1} \sum_{k=0}^{K} \gamma_k^\star \alpha_{m,k} e^{\frac{-j2\pi n \tau_k^\star}{NT_s}} \boldsymbol{A}_{N_R,N_T}\left(\theta_{Rx,k}^\star, \phi_{Tx,m}^{(n)}\right), \quad (13)$$

where $\alpha_{m,k} \doteq g_{N_T}\left(\frac{2m-M}{M} - \sin\theta_{Tx,k}^\star\right)$ and the function $g_{N_0}(\omega) \doteq \frac{1-e^{j\pi N_0 \omega}}{1-e^{j\pi\omega}}$, $\forall N_0 \in \mathbb{Z}\setminus\{0\}$ and $\omega \in (-1,1]$.

IV. Precoder Analysis

A. Approximate Spoofed Parameters at Eve

Herein, we derive an approximation on Eve's parameter estimates when the DSAS precoder is employed. The approximation is instrumental in predicting Eve's performance. We assume that Eve can accurately separate the channel contributions associated with each path, which are denoted by $\widetilde{\boldsymbol{H}}_k^{(n)}$ for the k-th propagation path. In this case, the channel contribution takes the form

$$\widetilde{\boldsymbol{H}}_k^{(n)} = \gamma_k^\star e^{\frac{-j2\pi n\tau_k^\star}{NT_s}} \sum_{m=0}^{M-1} \alpha_{m,k} \boldsymbol{A}_{N_R,N_T}\left(\theta_{Rx,k}^\star, \phi_{Tx,m}^{(n)}\right). \quad (14)$$

Then Eve's task is simplified to estimating $\theta_{Rx,k}$ and $\theta_{Tx,k}$ from $\widetilde{\boldsymbol{H}}_k^{(n)}$, $\forall k \in \{0,\cdots,K\}$. This can be framed as

$$\hat{\theta}_{Rx,k} = \arg\max_{\theta_{Rx,k}} \left\| \widetilde{H}_k^{(n)} \left(\widetilde{H}_k^{(n)}\right)^{\mathsf{H}} \boldsymbol{a}_{N_R}\left(\theta_{Rx,k}\right) \right\|_2^2 \quad (15a)$$

$$\hat{\theta}_{Tx,k} = \arg\max_{\theta_{Tx,k}} \left\| \left(\widetilde{H}_k^{(n)}\right)^{\mathsf{H}} \widetilde{H}_k^{(n)} \boldsymbol{a}_{N_T}\left(\theta_{Tx,k}\right) \right\|_2^2. \quad (15b)$$

By plugging Equations (2) and (14) into the objective function of Equation (15a), we have,

$$\left\| \widetilde{H}_k^{(n)} \left(\widetilde{H}_k^{(n)}\right)^{\mathsf{H}} \boldsymbol{a}_{N_R}\left(\theta_{Rx,k}\right) \right\|_2 = \left|\gamma_k^\star\right| N_R \left\|\boldsymbol{b}^{(n)}\right\|^2 \left|\boldsymbol{a}_{N_R}^{\mathsf{H}}\left(\theta_{Rx,k}^\star\right) \boldsymbol{a}_{N_R}\left(\theta_{Rx,k}\right)\right|, \quad (16)$$

where $\boldsymbol{b}^{(n)} \doteq \sum_{m=1}^{M-1} \overline{\alpha}_{m,k} \boldsymbol{a}_{N_T}\left(\phi_{Tx,m}^{(n)}\right)$ is not a function of $\theta_{Rx,k}$. From (16), we see that the optimal solution to the program (15a) is $\hat{\theta}_{Rx,k} = \theta_{Rx,k}^\star$.

Similarly, plug Equations (2) and (14) into Equation (15b),

$$\begin{aligned} &\left(N_R\left|\gamma_k^\star\right|\right)^{-1} \left(\widetilde{H}_k^{(n)}\right)^{\mathsf{H}} \widetilde{H}_k^{(n)} \boldsymbol{a}_{N_T}\left(\theta_{Tx,k}\right) \\ &= \sum_{p=1}^{M-1}\sum_{m=1}^{M-1} \overline{\alpha}_{p,k}\alpha_{m,k} \boldsymbol{a}_{N_T}\left(\phi_{Tx,p}^{(n)}\right) \boldsymbol{a}_{N_T}^{\mathsf{H}}\left(\phi_{Tx,m}^{(n)}\right) \boldsymbol{a}_{N_T}\left(\theta_{Tx,k}\right) \\ &= \left[\sum_{m=0}^{M-1} \alpha_{m,k} g_{N_T}\left(\sin\phi_{Tx,k}^{(n)} - \sin\theta_{Tx,k}\right)\right] \boldsymbol{b}^{(n)}. \end{aligned} \quad (17)$$

Since $\boldsymbol{b}^{(n)}$ is not a function of $\theta_{Tx,k}$, finding the solution to the program (15b) is equivalent to solving

$$\max_{\theta_{Tx,k}} \left| \sum_{m=0}^{M-1} \alpha_{m,k} g_{N_T}\left(\sin\phi_{Tx,k}^{(n)} - \sin\theta_{Tx,k}\right) \right|. \quad (18)$$

Notice that the magnitude of the function $g_{N_0}(\omega)$ peaks at $\omega = 0$ and quickly diminishes away from $\omega = 0$. Therefore, the AoD estimate can be approximated as $\hat{\theta}_{Tx,k} \approx \phi_{Tx,m_k}^{(n)}$, where $m_k \doteq \arg\max_{0\le m\le M-1} |\alpha_{m,k}| = \arg\min_{0\le m\le M-1} \min_{t\in\mathbb{Z}} \left|2t + \frac{2m-M}{M} - \sin\theta_{Tx,k}^\star\right|$.

By properly choosing $\left\{\phi_{k,m_k}^{(n)}\right\}_{k=0}^{M}$, we can ensure that, almost surely, $\left|\phi_{Tx,m_k}^{(n)} - \phi_{Tx,m_0}^{(n)}\right| \neq \left|\theta_{Tx,k}^\star - \theta_{Tx,0}^\star\right|$, thereby misleading Eve into estimating the wrong ADoDs. The approximation on spoofed ADoDs enables an analysis on the localization error of Eve. Our precoder design is seen to be effective in the numerical results, as motivated by the analysis below.

B. Bias Calculation for Eve

For the analysis, we apply the same precoding matrix $\boldsymbol{\Psi}\left(\boldsymbol{\phi}_{Tx}^{(n)}\right) = \boldsymbol{\Psi}\left(\boldsymbol{\phi}_{Tx}\right)$ over all subcarriers. Eve will infer $\widetilde{\boldsymbol{\tau}} = \boldsymbol{\tau}^\star$, $\widetilde{\boldsymbol{\theta}}_{Rx} = \boldsymbol{\theta}_{Rx}^\star$ and $\widetilde{\boldsymbol{\theta}}_{Tx} = [\phi_{Tx,m_0},\cdots,\phi_{Tx,m_K}]^\top$ as the parameters of the spoofed channel, where we define $\dot{\phi}_{Tx,m_k} \doteq |\phi_{Tx,m_k} - \phi_{Tx,m_0}|$. While this is an idealized situation for Eve, in practice, the averaged bias is close to (19), as shown in our numerical results. Denote Eve's estimate of Alice's location as $\widetilde{\boldsymbol{p}}$. Given the differential localization in Equations (5), (8) and (9), the error between the spoofed and ground-truth locations is the error in the distance estimate,

$$\begin{aligned} \left\|\boldsymbol{p}^\star - \widetilde{\boldsymbol{p}}\right\|_2 &= \left| \sum_{k=1}^{K} c_k^\star d_k^\star - \sum_{k=1}^{K} \widetilde{c}_k \widetilde{d}_k \right| \overset{(c)}{=} \left| \sum_{k=1}^{K} \widetilde{c}_k d_k^\star - \sum_{k=1}^{K} \widetilde{c}_k \widetilde{d}_k \right| \\ &\overset{(d)}{=} \left| \sum_{k=1}^{K} \frac{\widetilde{c}_k c \dot{\tau}_k^\star}{2} \cot\frac{\dot{\theta}_{Rx,k}^\star}{2} \left(\cot\frac{\dot{\theta}_{Tx,k}^\star}{2} - \cot\frac{\dot{\phi}_{Tx,m_k}}{2}\right) \right|. \end{aligned} \quad (19)$$

where (c) is due to $\sum_{k=1}^{K} c_k^\star = \sum_{k=1}^{K} \widetilde{c}_k = 1$ and $d_k^\star = d^\star$, for $k = 1,\cdots,K$. Equality (d) follows by plugging in Equation (9).

From Equation (19), we observe that a non-zero localization bias exists under the spoofed parameters.

We assume ϕ_{Tx,m_k} are randomly sampled from the set $\left\{\sin^{-1}\left(\frac{2m+1-M_\phi}{M_\phi}\right) : m = 0,\cdots,M_\phi-1\right\}$ without replacement; this ensures that $\boldsymbol{\Psi}\left(\boldsymbol{\phi}_{Tx}^{(n)}\right)$ is unitary when $M_\phi = M$. Let $K = 1$ in Equation (19). By defining $K_0 \doteq \frac{c\dot{\tau}_1^\star}{2}\cot\frac{\dot{\theta}_{Rx,1}^\star}{2}$,

$$\begin{aligned} \mathbb{E}\left[\left\|\boldsymbol{p}^\star - \widetilde{\boldsymbol{p}}\right\|_2\right] &\overset{(e)}{\geq} K_0 \left[\mathbb{E}\left[\cot\frac{\dot{\phi}_{Tx,1}}{2}\right] - \cot\frac{\dot{\theta}_{Tx,1}^\star}{2}\right] \\ &\overset{(f)}{\geq} K_0 \left[\left(\frac{M_\phi}{2M_\phi - 1} H_{M_\phi - 1} - 2\right) - \cot\frac{\dot{\theta}_{Tx,1}^\star}{2}\right] \\ &\overset{(g)}{\geq} K_0 \left[\Theta\left(\log M_\phi\right) - \cot\frac{\dot{\theta}_{Tx,1}^\star}{2}\right] = \Theta\left(\log M_\phi\right), \end{aligned} \quad (20)$$

where (e) follows from Jensen's inequality and $|x| \geq x, \forall x \in \mathbb{R}$. Inequality (f) holds by the inequality $\cot\left|\frac{\vartheta-\varphi}{2}\right| \geq \frac{2-|\sin\vartheta|-|\sin\varphi|}{|\sin\vartheta-\sin\varphi|}$ for $\vartheta,\varphi \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, and under the assumption of randomly sampling $\sin\phi_{Tx,m_0} \neq \sin\phi_{Tx,m_1}$ from $\left\{\frac{2m+1-M_\phi}{M_\phi} : m = 0,\cdots,M_\phi-1\right\}$, where $\mathcal{H}_n \doteq \sum_{i=1}^{n}\frac{1}{i}$. (g) follows from the n-th harmonic number $\mathcal{H}_n = \Theta(\log n)$ [21].

To increase the spoofing error at $K = 1$, Equation (20) suggests that $M_\phi \to \infty$, which is equivalent to drawing $\sin\phi_{Tx,k} \sim \text{Uniform}(-1, 1)$. However, Bob's performance degrades as M_ϕ increases in practice. Therefore, a trade-off exists in choosing M_ϕ: an effective choice is $M_\phi = M = N_T$.

V. Simulation Results

The following parameters are fixed: $N = 81$, $N_R = N_T = 8$, $B = 30$MHz. For each channel realization, generate $K+1$ i.i.d. random variables $\tau_k^\star \sim \text{Uniform}(0, NT_s)$, which are relabeled in ascending order as $\tau_0^\star \le \cdots \le \tau_K^\star$; generate $\theta_{Tx,k}^\star \sim \text{Uniform}\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$; place Eve/Bob at $[0,0]^\top$, and compute Alice's location and AoAs as $\boldsymbol{p}^\star = -c\tau_0^\star[\cos\theta_{Tx,0}, \sin\theta_{Tx,0}]^\top$ and $\theta_{Rx,k}^\star \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, respectively; generate $\gamma_k^\star \sim \mathcal{CN}\left(0, \exp\left\{-0.05\frac{\tau_k^\star}{NT_s}\right\}\right)$. For DSAS, set $M = N_T = 8$ and $\boldsymbol{\phi}_{Tx}^{(n)} = \boldsymbol{\phi}_{Tx}$ for $n = 1, \cdots, N$, and generate the spoofed AoDs $\boldsymbol{\phi}_{Tx}$ from the set $\left\{\sin^{-1}\left(\frac{2m+1-M_\phi}{M_\phi}\right) : m = 0, \cdots, M_\phi - 1\right\}$ without replacement. For DAIS, we adapt the precoding matrix $\boldsymbol{\Phi}^{(n)}(\Delta_\tau = 0.5NT_s, \Delta_\theta = 0.25\pi)$ according to (10). For DAIS+DSAS, we use the product of the precoders as $\boldsymbol{\Phi}^{(n)}(0.5NT_s, 0.25\pi)\,\boldsymbol{\Psi}(\boldsymbol{\phi}_{Tx})$ on the n-th subcarrier. We compare the performances of above spoofing strategies under the non-differential (ND) localization method used in [16].

The evaluation metric is : NRMSE $= \mathbb{E}\left[\frac{\|\boldsymbol{p}^\star - \hat{\boldsymbol{p}}\|_2}{\|\boldsymbol{p}^\star - \boldsymbol{q}^\star\|_2}\right]$, where the expectation is taken w.r.t. channel, precoder and noise realizations. Let $\boldsymbol{W} \in \mathbb{C}^{N_R \times N_R}$ and $\boldsymbol{F} \in \mathbb{C}^{N_T \times N_T}$ be unitary matrices, and assume all symbols have unit energy, i.e. $\left|S_{r_1,r_2}^{(n)}\right| = 1$. The signal-to-noise ratio is defined as $\text{SNR(dB)} = 10\log_{10}\left[\sum_{n=1}^{N}\left\|\boldsymbol{H}^{(n)}\right\|_F^2 / \left(NN_RN_T\sigma^2\right)\right]$, where $\boldsymbol{H}^{(n)}$ is replaced by $\widetilde{\boldsymbol{H}}^{(n)}$ for spoofed channels. We estimate the parameters using FLEX [4] with correlation-based angle estimation. We assume that the number of paths is unknown and all the paths have distinct AoAs and AoDs.

A. Impact of SNR(dB)

Figure 1 shows that Bob's performance overlaps with that under non-spoofing channels, verifying that the spoofing effect can be eliminated with the side information. The bias introduced by DAIS and DSAS dominates Eve's errors, causing Eve's curves to appear nearly flat. As expected, DAIS introduces localization errors greater than 0.033 for Eve in only around 74% of 220 channel realizations, whereas the normalized localization errors produced by DSAS and DAIS+DSAS exceed 0.4 for all channel realizations. By combining the two strategies, DAIS+DSAS achieves the best overall performance, increasing Eve's localization error by approximately 8dB compared with DAIS. Under the ND localization method (dashed lines), DSAS and DAIS+DSAS schemes are still effective.

B. Impact of M_ϕ

In Figure 2, we investigate into the impact of the minimum separation between the sine of spoofing AoDs, $\frac{2}{M_\phi}$, on the spoofing performance. The error predicted by Equation (19)

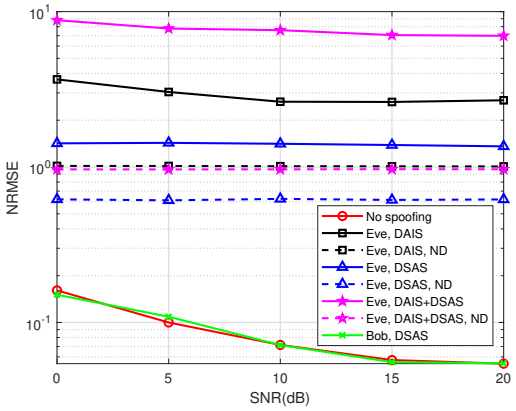


Figure 1. NRMSE results versus SNR(dB). Set $M_\phi = 8$, $K = 2$.

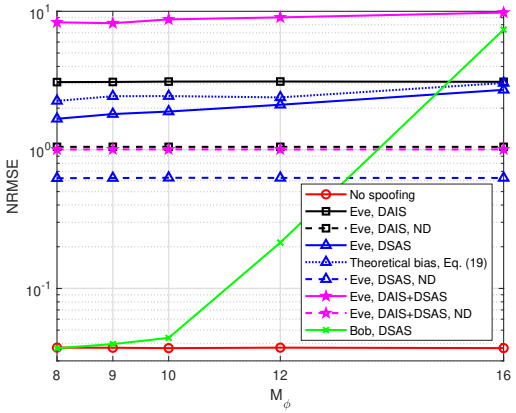


Figure 2. NRMSE results versus M_ϕ. Set SNR $= 20$dB, $K = 2$.

approaches Eve's performance under DSAS. As indicated by (20), the averaged error introduced by DSAS increases slowly with M_ϕ. However, as M_ϕ increases, Bob's performance degrades rapidly, because the condition number of $\boldsymbol{\Psi}\left(\boldsymbol{\phi}_{Tx}^{(n)}\right)$ increases, resulting in noise amplification when Bob reverses the spoofing effect. It suggests that we should choose $M_\phi = M = N_T$, where $\boldsymbol{\Psi}\left(\boldsymbol{\phi}_{Tx}^{(n)}\right)$ becomes a unitary matrix.

VI. Conclusions

In this work, we proposed a CSI-free precoding matrix design, which distorts Eve's channel estimation, while allowing Bob to reverse the spoofing effect. By applying the DSAS precoder, we analyzed the channel parameters that Eve infers from the spoofed channel, demonstrating that DSAS produces spoofed ADoDs for Eve. This confirms the effectiveness of DSAS under the differential-signaling localization, where DAIS will fail unless phase wrapping occurs. With the approximate spoofed parameters, we derived the localization bias introduced into Eve's location estimate. Simulation results verify that the DSAS strategy introduces a significant localization bias into Eve's estimates. When combined with DAIS, DSAS achieves the best overall spoofing performance.

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REFERENCES

[1] T. S. Rappaport et al., "Millimeter wave mobile communications for 5g cellular: It will work!", *IEEE access*, vol. 1, pp. 335–349, 2013.

[2] A. Fascista, A. Coluccia, H. Wymeersch, and G. Seco-Granados, "Downlink single-snapshot localization and mapping with a single-antenna receiver", *IEEE Transactions on Wireless Communications*, vol. 20, no. 7, pp. 4672–4684, 2021.

[3] J. Li, M. F. Da Costa, and U. Mitra, "Joint localization and orientation estimation in millimeter-wave mimo ofdm systems via atomic norm minimization", *IEEE Transactions on Signal Processing*, vol. 70, pp. 4252–4264, 2022.

[4] A. Pourafzal, H. Huang, V. Pettersson, M. F. Keskin, and H. Wymeersch, "Flex: Low-complexity 5d beamspace channel estimation for mmwave mimo-ofdm", in *2025 33rd European Signal Processing Conference (EUSIPCO)*, IEEE, Palermo, Italy, 2025, pp. 920–924.

[5] S. Goel and R. Negi, "Guaranteeing secrecy using artificial noise", *IEEE transactions on wireless communications*, vol. 7, no. 6, pp. 2180–2189, 2008.

[6] S. Tomasin, "Beamforming and artificial noise for cross-layer location privacy of e-health cellular devices", in *2022 IEEE International Conference on Communications Workshops (ICC Workshops)*, IEEE, Seoul, Korea, Republic of, 2022, pp. 568–573.

[7] J. J. Checa and S. Tomasin, "Location-privacy-preserving technique for 5g mmwave devices", *IEEE Communications Letters*, vol. 24, no. 12, pp. 2692–2695, 2020.

[8] R. Ayyalasomayajula, A. Arun, W. Sun, and D. Bharadia, "Users are closer than they appear: Protecting user location from wifi aps", in *Proceedings of the 24th International Workshop on Mobile Computing Systems and Applications*, 2023, pp. 124–130.

[9] Y. Zhang et al., "Privacy preservation in mimo-ofdm localization systems: A beamforming approach", *IEEE Wireless Communications Letters*, vol. 14, no. 7, pp. 1979–1983, 2025.

[10] Y. Zhu and J. Wang, "Securing wireless localization: The role of power allocation in location privacy protection", *IEEE Transactions on Communications*, vol. 73, no. 12, pp. 15 639–15 655, 2025.

[11] M. Srinivasan, L. Senigagliesi, H. Chen, A. Chorti, M. Baldi, and H. Wymeersch, "Aoa-based physical layer authentication in analog arrays under impersonation attacks", in *2024 IEEE 25th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, IEEE, Lucca, Italy, 2024, pp. 496–500.

[12] P. Maity et al., "From pilot to precoding design: Blind angular spoofing for location privacy in mimo systems", *arXiv preprint arXiv:2605.00535*, 2026.

[13] A. Arun, V. Anand, W. Sun, R. Ayyalasomayajula, and D. Bharadia, "Dolos: Tricking the wi-fi aps with incorrect user locations", *arXiv preprint arXiv:2407.16138*, 2024.

[14] J. Li and U. Mitra, "Channel state information-free location-privacy enhancement: Delay-angle information spoofing", in *ICC 2024-IEEE International Conference on Communications*, IEEE, 2024, pp. 3767–3772.

[15] J. Li and U. Mitra, "Optimized parameter design for channel state information-free location spoofing", in *2024 IEEE International Symposium on Information Theory (ISIT)*, IEEE, Athens, Greece, 2024, pp. 3695–3700.

[16] J. Li and U. Mitra, "Delay-angle information spoofing for channel state information-free location-privacy enhancement", *arXiv preprint arXiv:2504.14780*, 2025.

[17] J. Palacios, N. González-Prelcic, and C. Rusu, "Low complexity joint position and channel estimation at millimeter wave based on multidimensional orthogonal matching pursuit", in *2022 30th European Signal Processing Conference (EUSIPCO)*, IEEE, Belgrade, Serbia, 2022, pp. 1002–1006.

[18] A. Arun, S. Saruwatari, S. Shah, and D. Bharadia, "Xrloc: Accurate uwb localization to realize xr deployments", in *Proceedings of the 21st ACM Conference on Embedded Networked Sensor Systems*, Istanbul, Turkiye, 2023, pp. 459–473.

[19] M. R. Shamsian, M. Sadeghi, and F. Behnia, "Joint tdoa and doa single site localization in nlos environment using virtual stations", *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–10, 2023.

[20] Y. Chen, N. González-Prelcic, T. Shimizu, and H. Lu, "Learning to localize with attention: From sparse mmwave channel estimates from a single bs to high accuracy 3d location", *arXiv preprint arXiv:2307.00167*, 2023.

[21] B.-N. Guo and F. Qi, "Sharp bounds for harmonic numbers", *Applied Mathematics and Computation*, vol. 218, no. 3, pp. 991–995, 2011.